

## Introduction

Many engineering students struggle to connect physical strain measurements to stress-state visualization in real time.

This project addresses this gap by developing a handheld system that measures strain and visualizes stress states and Mohr's Circle through an integrated hardware-software interface.



Low system latency  
( $\leq 100$  ms)



User-friendly interface for real-time stress interpretation



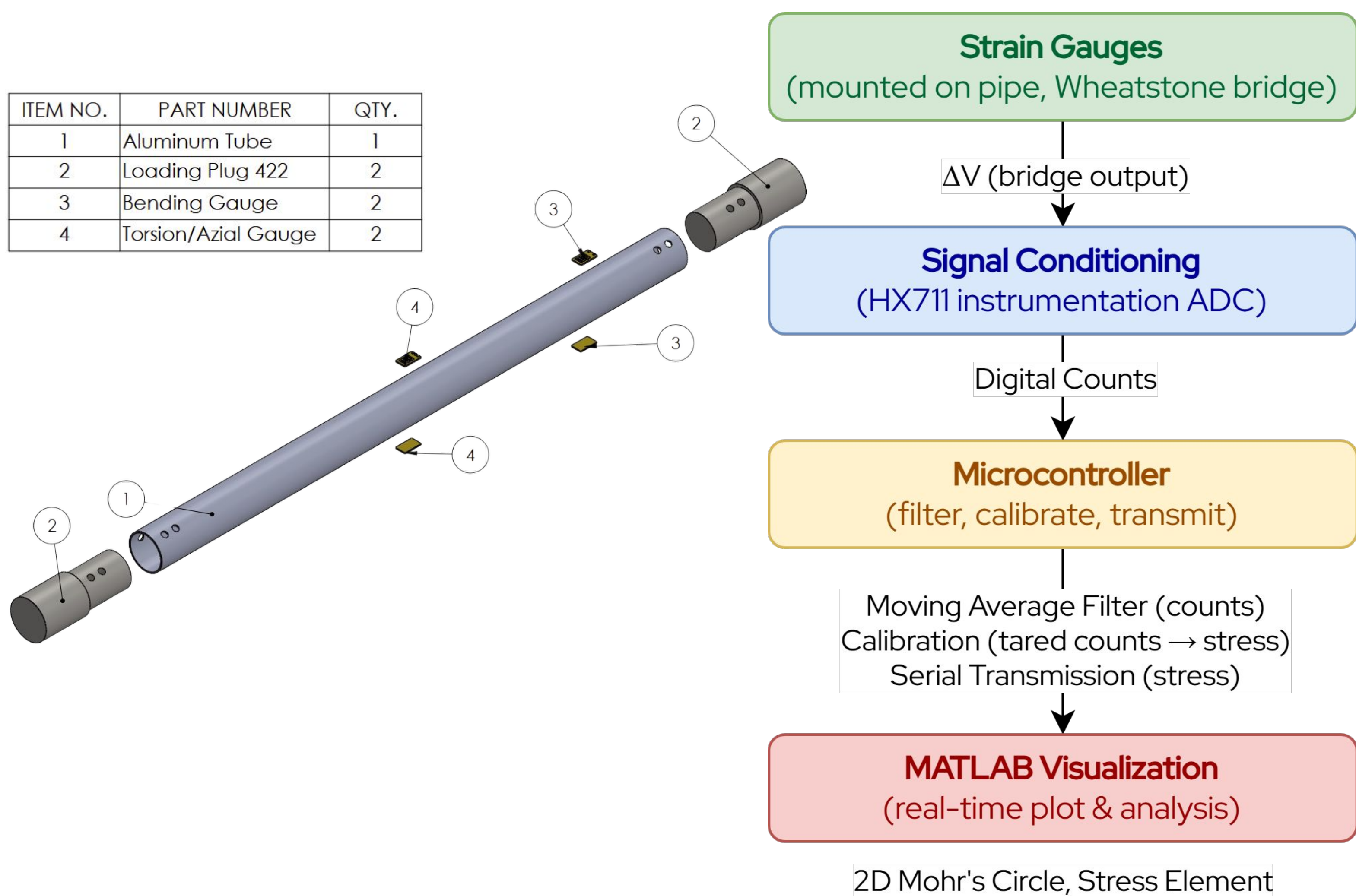
Robust signal processing



Minimum strain resolution ( $\leq 1 \mu\epsilon$ )

## System Overview

| ITEM NO. | PART NUMBER         | QTY. |
|----------|---------------------|------|
| 1        | Aluminum Tube       | 1    |
| 2        | Loading Plug 422    | 2    |
| 3        | Bending Gauge       | 2    |
| 4        | Torsion/Axial Gauge | 2    |

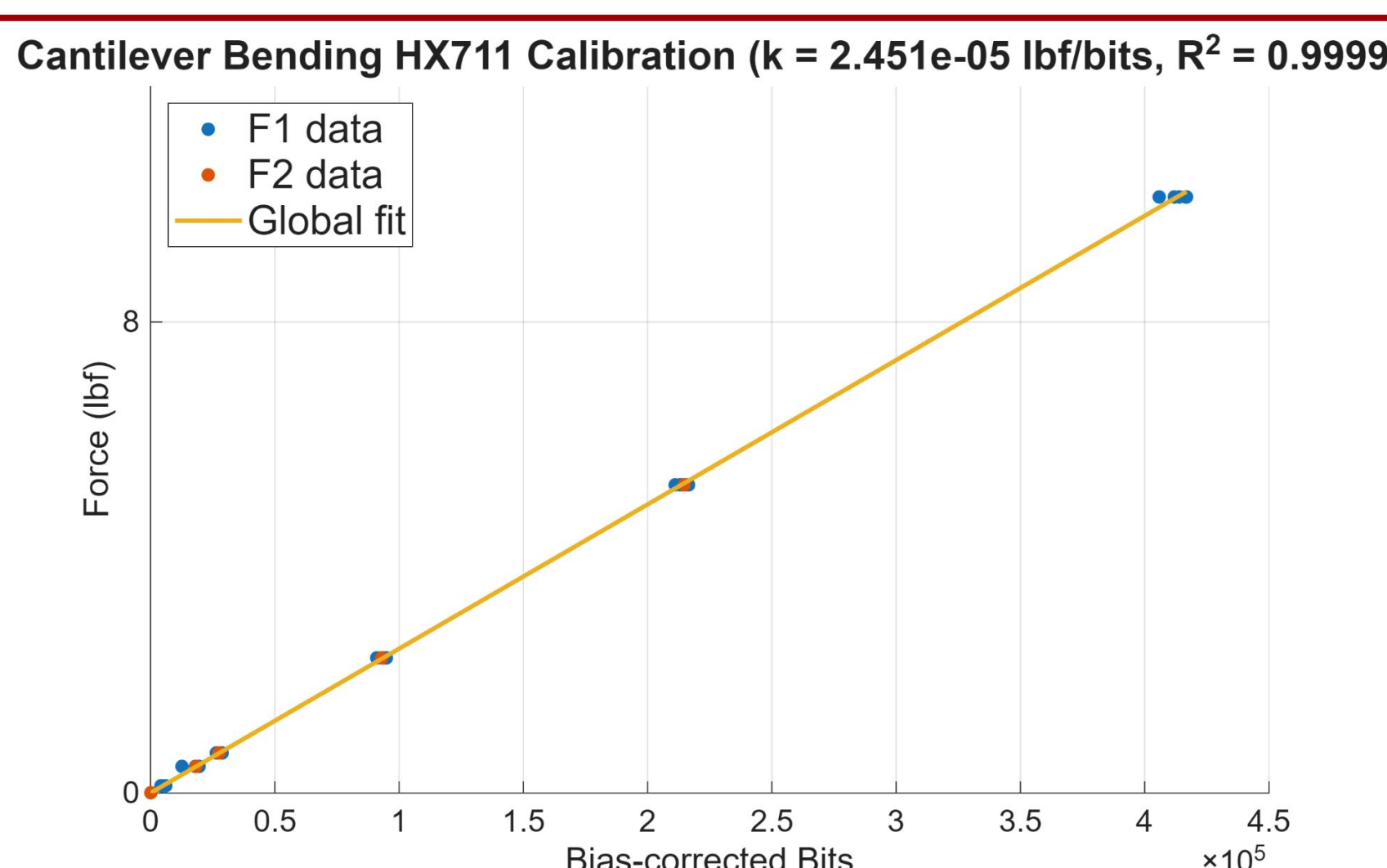
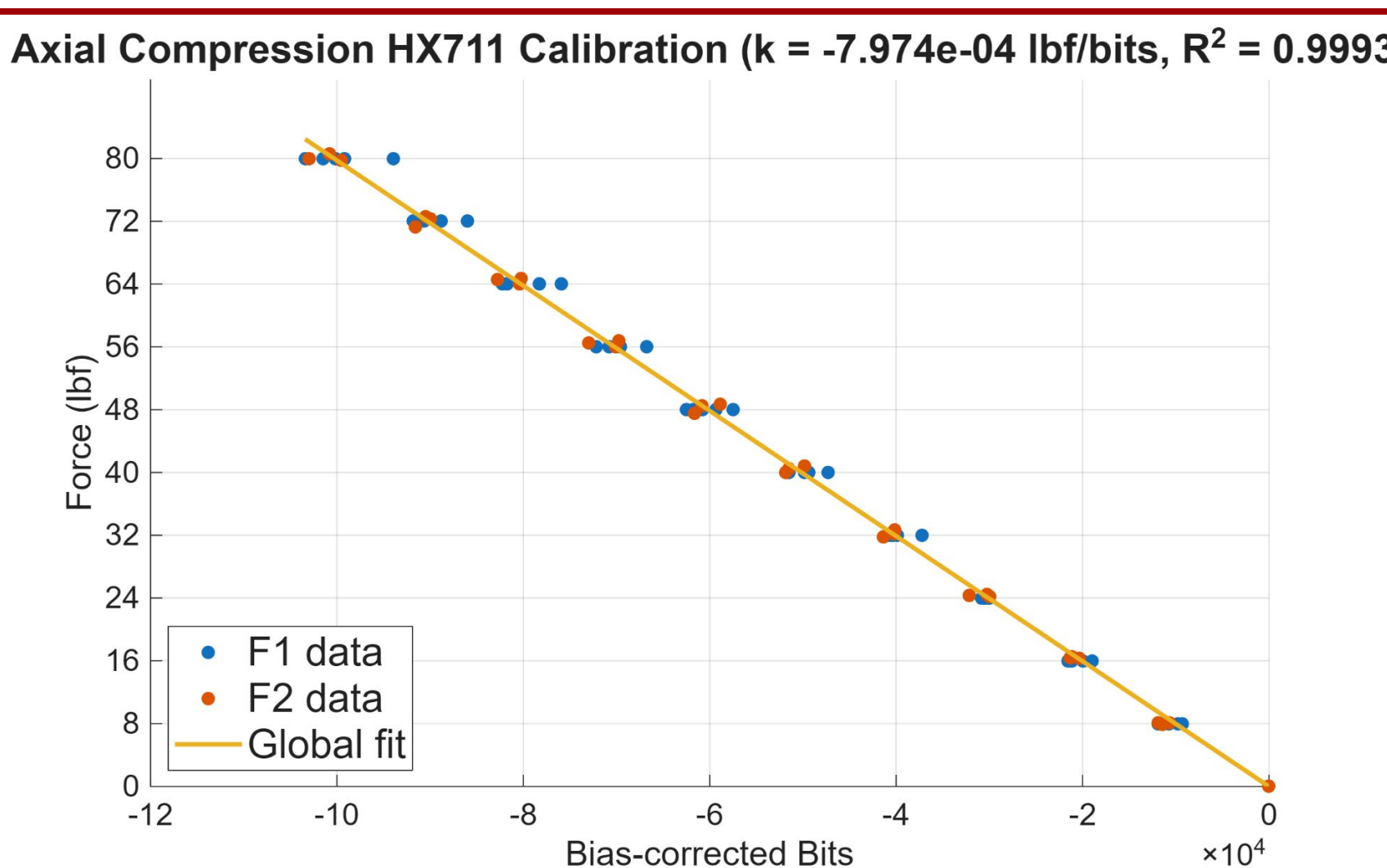


## Validation & Results

### Estimation of Effective Resolution

- Datasheet RMS ( $1\sigma$ ) resolution is roughly 33 nV
- Experimental RMS ( $1\sigma$ ) resolution is roughly 136 nV
- Practical noise-free strain resolution ( $3\sigma$ ) is  $0.124 \mu\epsilon$

$$\epsilon_{noise} = \frac{V_{noise} \cdot B}{V_{ex} \cdot GF} \quad V_{noise} = \frac{\sigma_N}{2^{23}} \cdot \frac{V_{ex}}{G}$$



| Parameter                | Value        |
|--------------------------|--------------|
| $V_{ex}$                 | 3.28 V       |
| GF                       | $\sim 2.0$   |
| G                        | 128          |
| B (half-bridge)          | 2            |
| $\sigma_N$ (500 samples) | 44.69 counts |

### Bit-to-Force Calibration and Validation

#### Experimental setup

- Axial loading: 0-80 lbf compression
- Bending: 0-236 in-lbf cantilever, end load
- HX711 ADC, 10 SPS, 300-sample batch
- $\geq 5$  loading cycles per case

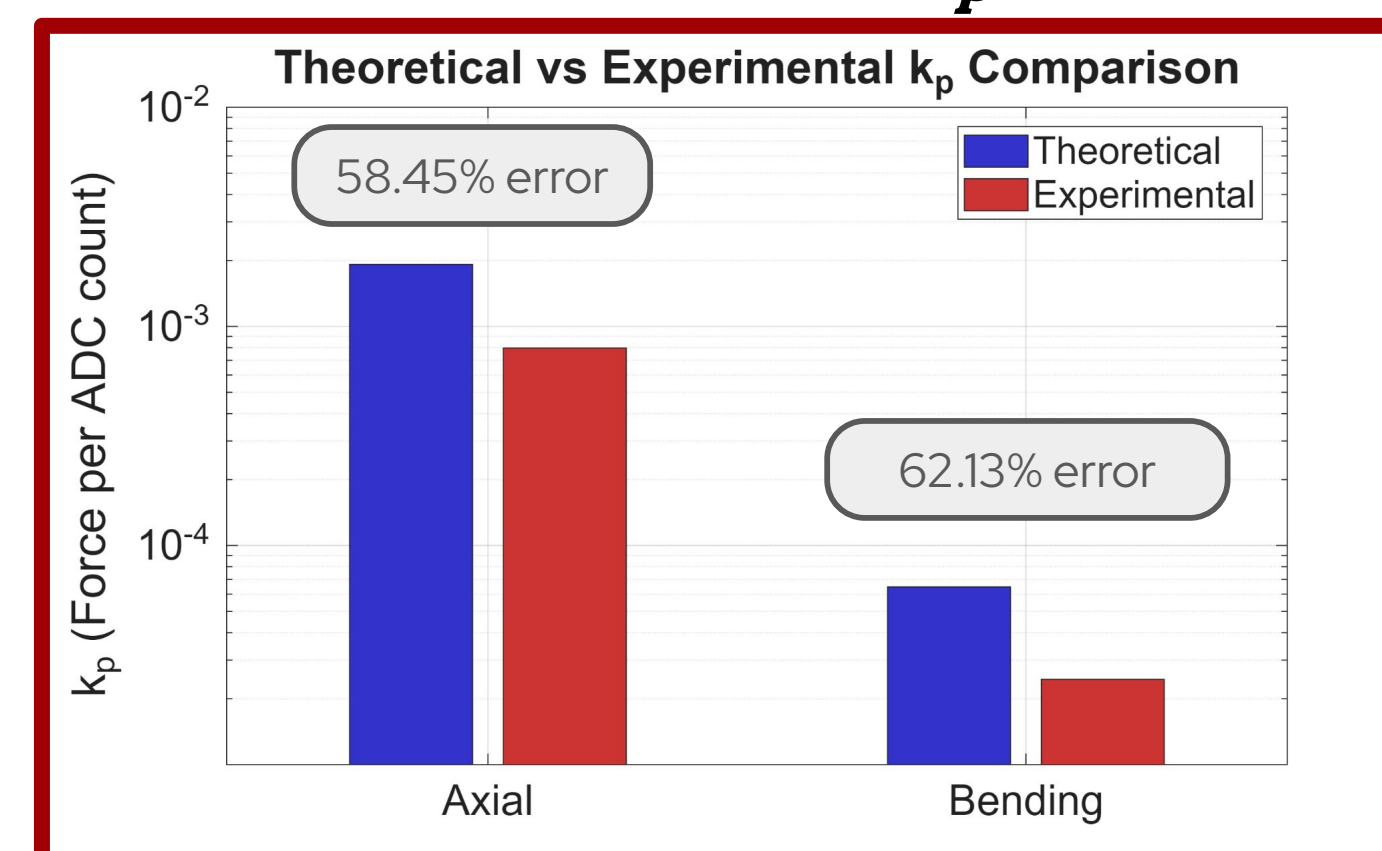
#### Post-processing and calibration

- Zero-offset load removed from raw ADC counts
- Linear regression to compute bit-to-force scalar  $k_p$

#### Validation

- Experimental scalar compared against theoretical model

### Calibration Constant ( $k_p$ ) Validation



$$\% \text{ error} = \frac{|k_p^{\text{exp}} - k_p^{\text{theory}}|}{|k_p^{\text{theory}}|} \times 100\%$$

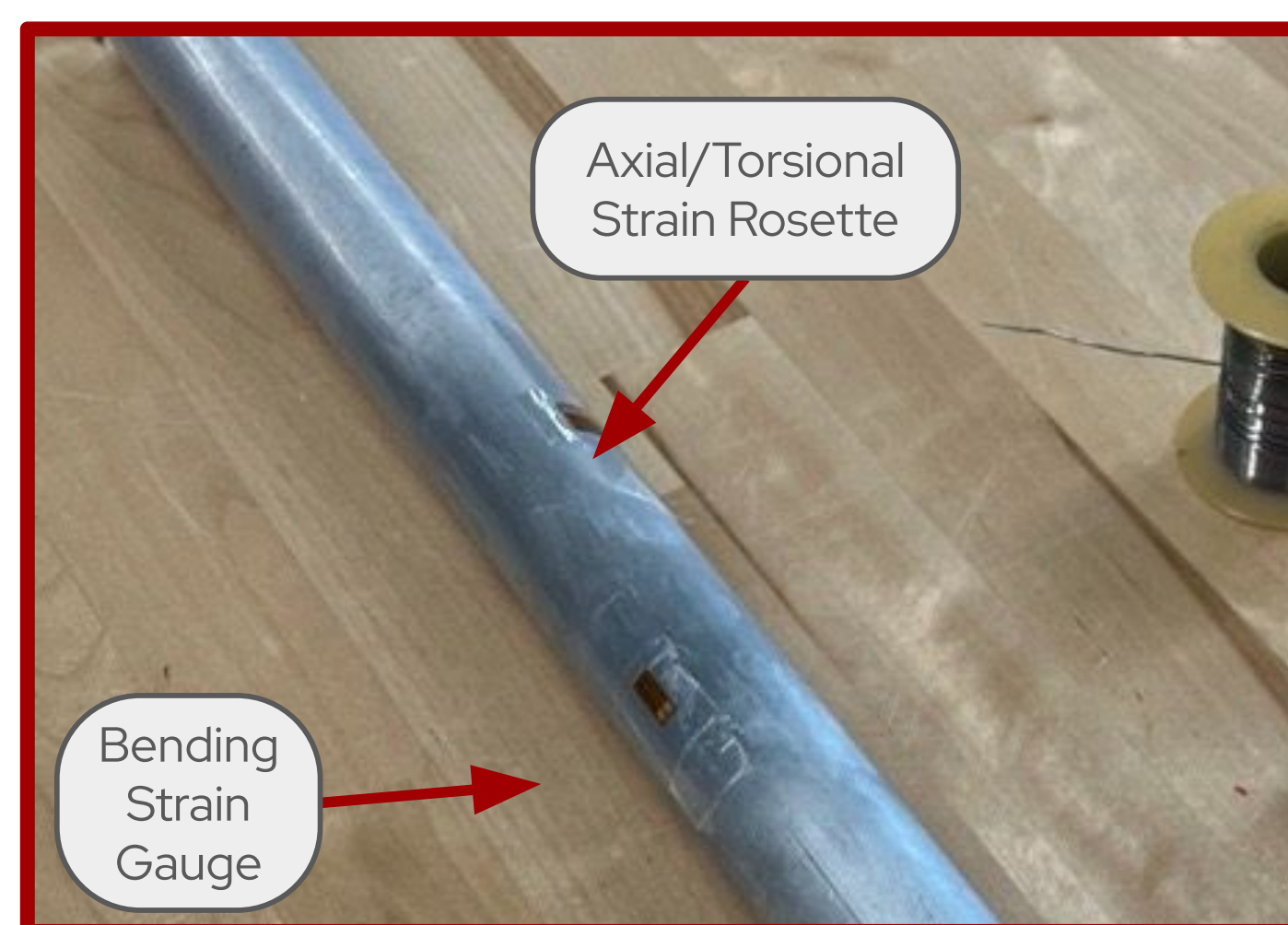
### Theoretical $k_p$ Derivation

- G: amplifier gain
- GF: gauge factor
- B: bridge factor
- $k_p/k_\epsilon$ : bits-to-force/strain conversion factor

$$k_\epsilon = \frac{B}{G \cdot GF \cdot 2^{23}}$$

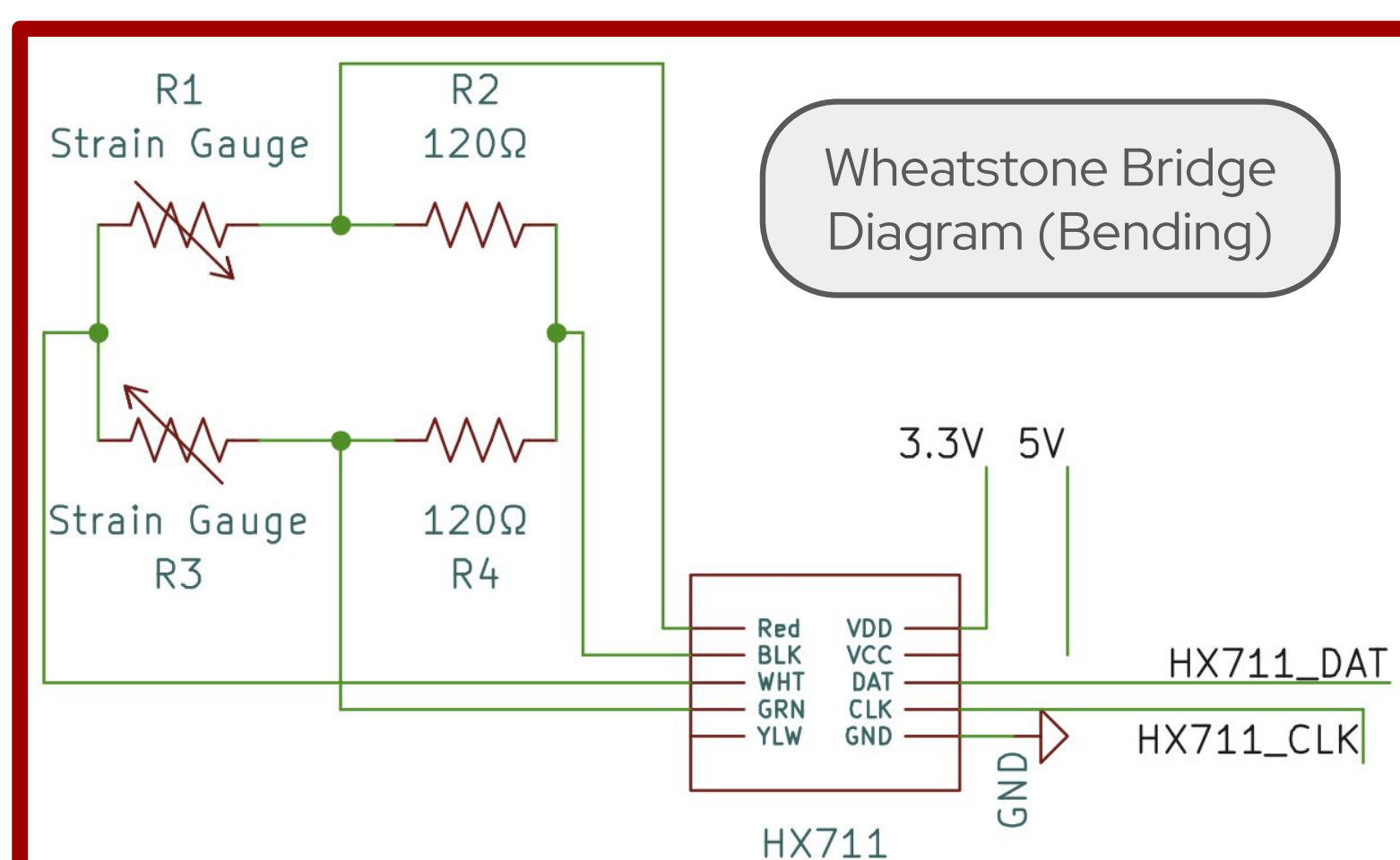
$$\begin{cases} k_p^{\text{axial}} = E \cdot A \cdot k_\epsilon \\ k_p^{\text{bending}} = \frac{E \cdot I \cdot k_\epsilon}{(L-x)y} \end{cases}$$

## Technical Design



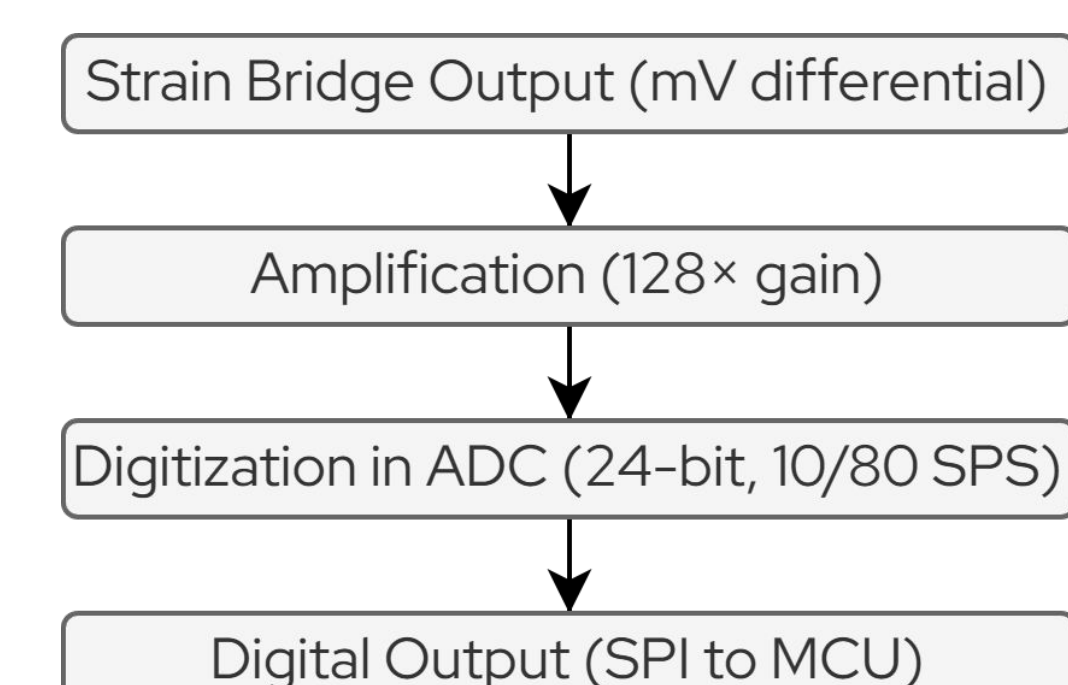
### Mechanical Design

- Minimum of two gauges per bridge to double sensitivity and compensate for thermal drift
- Axial/torsional strain rosettes and bending gauges positioned on the neutral axis of bending to isolate torsional response
- Bending gauges placed at center of pipe to maximize strain under symmetric loading



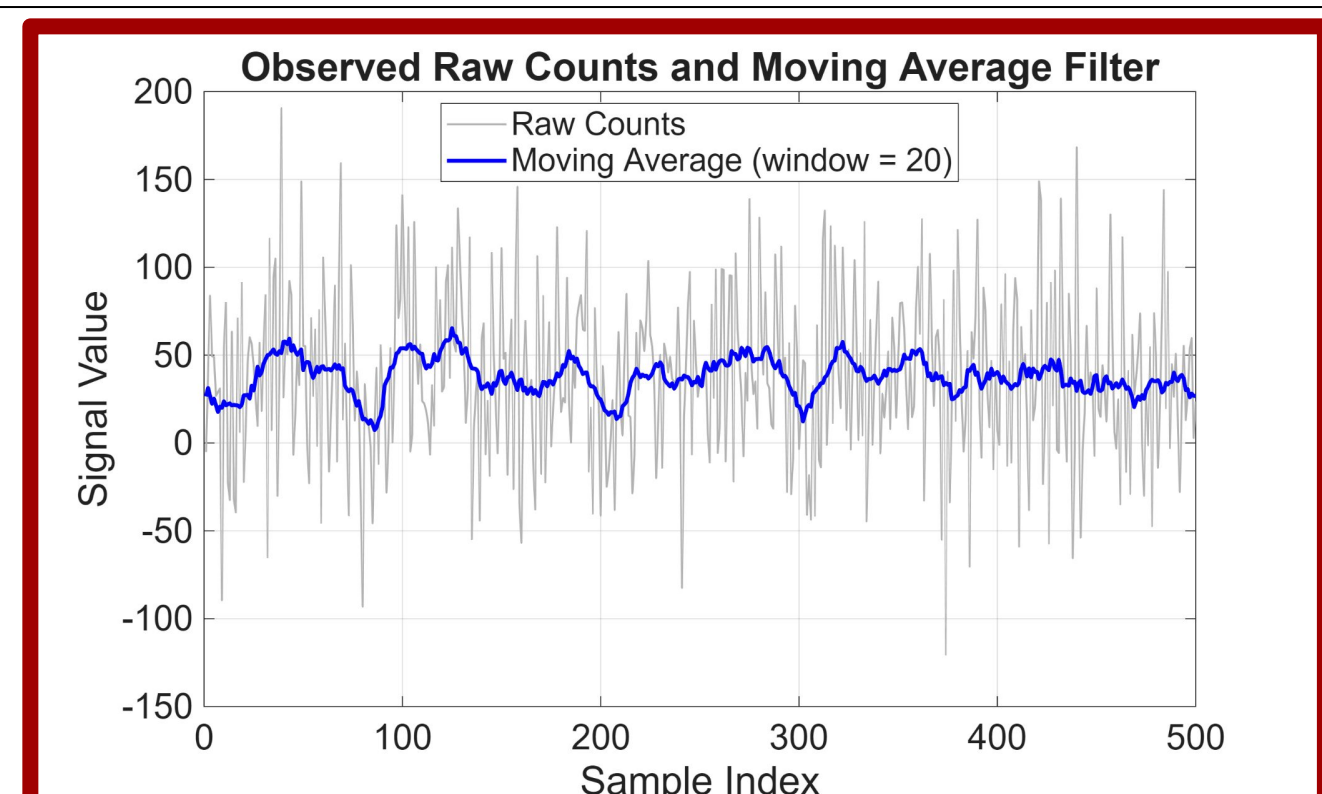
### Wheatstone Bridge Design

- Half-bridge configuration: R1 & R3 are active strain gauges, doubling sensitivity
- R2 & R4 are fixed 120Ω resistors reducing differential output under no load
- Bridge output feeds directly into HX711 differential inputs



### HX711 Amplification and Signal Conditioning

- 24-bit ADC ( $\sim 18$ -bit noiseless) and  $128\times$  programmable gain provides theoretical strain resolution of  $0.05$ - $0.1 \mu\epsilon$
- 10 samples/s output rate selected over 80 samples/s to achieve a higher noiseless resolution

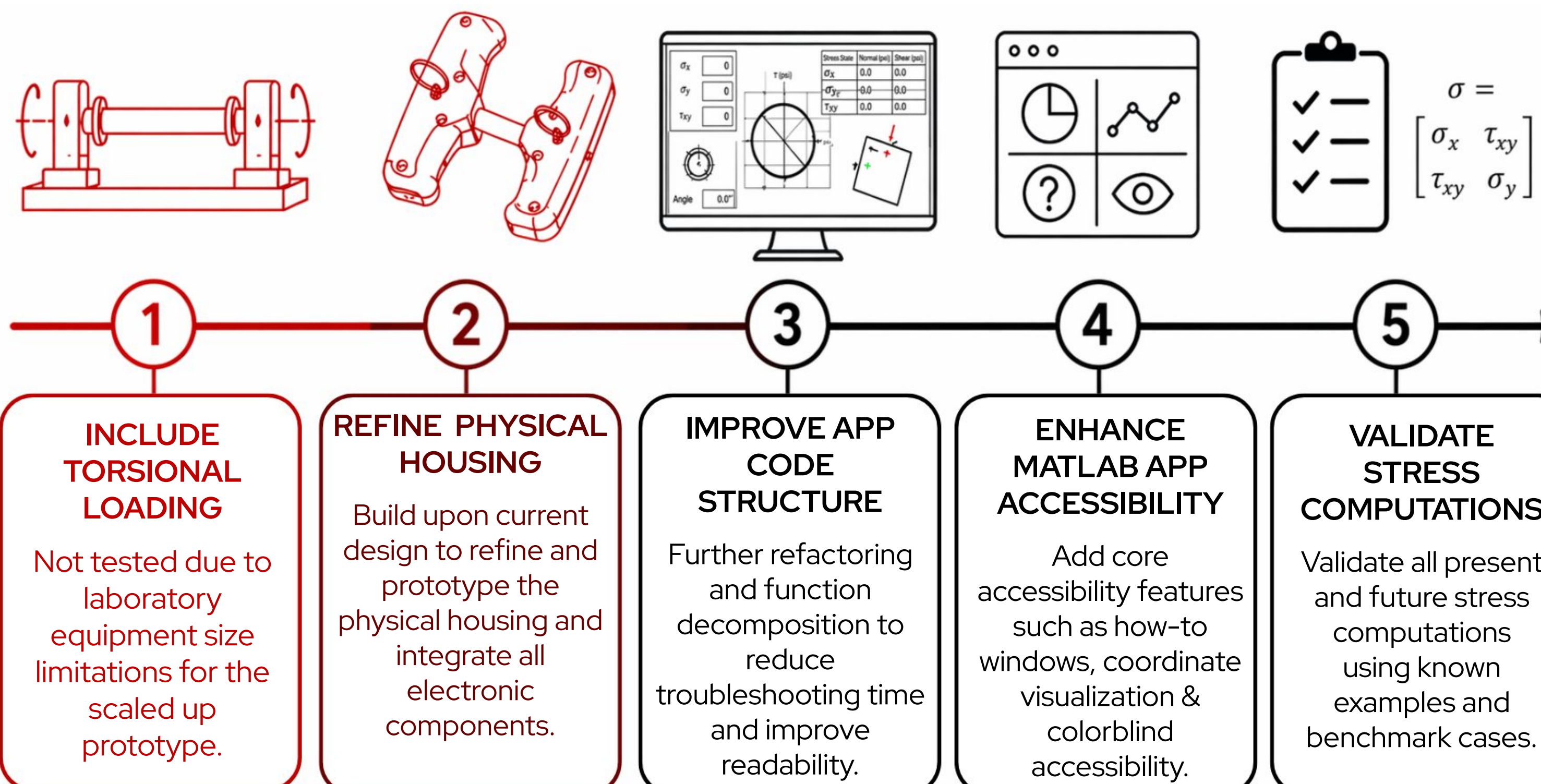


### Arduino Signal Processing

- Uses zero-offset calibration to remove unloaded bridge bias from raw HX711 counts
- Takes moving average of signal to reduce high-frequency noise
- Converts filtered ADC counts to engineering stress units (MPa)

## Future Work

Future work by the next team will continue testing and refining the physical integration of the device by:



## References

- [1] H. Qi, O. Liu, Y. Gan, and W. Dong, "Work-in-Progress: Learning Tool to Enhance Understanding of Stress States and Mohr's Circle," ASEE Annual Conference & Exposition, 2024
- [2] S. Livingston-Jha, H. Deng, Y. Huai, and J. Detlor, "Work-in-Progress: Further Work on a Learning Tool to Enhance Understanding of Stress States and Mohr's Circle," ASEE Annual Conference & Exposition, 2025.
- [3] J. C. Moller and A. Mokaddem, "A Tool for Teaching Stress Transformation by Mohr's Circle," in ASEE Annual Conference & Exposition, 1997.